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GRADE A+
ACCREDITED UNIVERSITY

Bachelor of Computer Applications

Semester – V

Business Analytics

Experiment No.: 2

**Title: Sales Performance Analysis Across Products, Categories,
and Regions using Power BI Visualizations**

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Sales Performance Analysis Across Products, Categories, and Regions using Power BI Visualizations

Github Link: <https://github.com/keerthi19hub/Business-Analytics>

Problem Statement

A retail company wants to understand how its sales and profit are doing across different products, categories, and regions. As the analyst, my job was to take the Sample Superstore dataset, clean it up in Power BI, and build a dashboard that makes these patterns easy to see.

Step 1: Getting the Data In

I loaded the Sample_Superstore data into Power BI as a table. It has close to 10,000 rows, with columns like Order Date, Ship Date, Customer ID, Segment, Region, City, State, Product ID, Category, and Sub-Category.

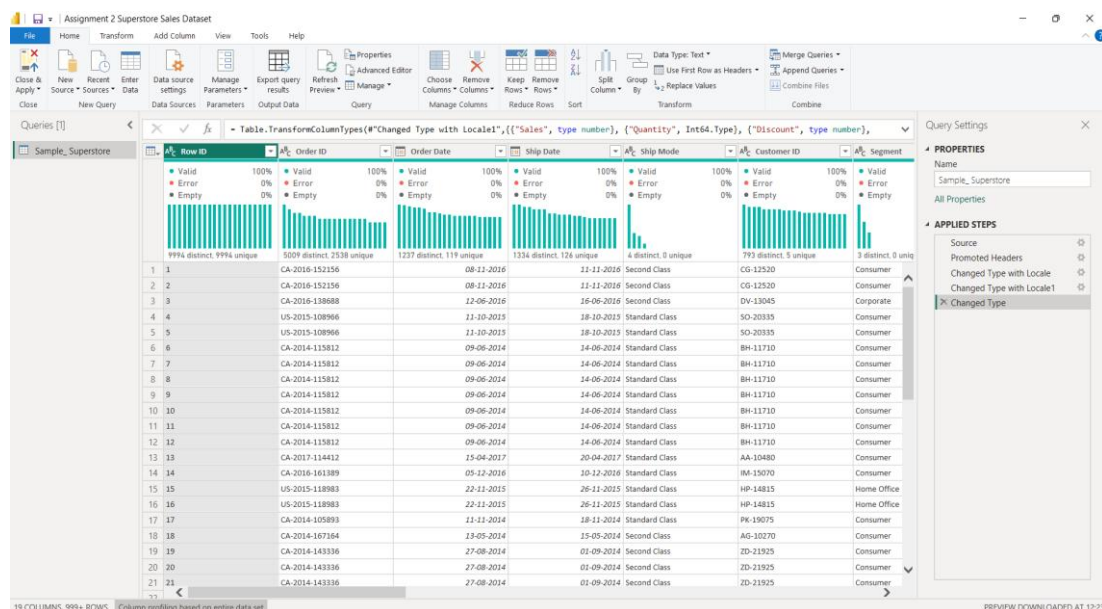
- The table itself contains 9,994 rows, so it's a decent-sized dataset to work with.
- Each row is one order line, meaning a single order can appear more than once if it had multiple products.
- Before doing anything else, I scrolled through the data a bit just to get a feel for what kind of values were in each column.

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Segment	Country	City	State	Region	Product ID	Category	Sub-Category
65	CA-2015-135545	24 November 2015	30 November 2015	Standard Class	KM-16720	Consumer	United States	Los Angeles	California	West	OFF-PA-10003892	Office Supplies	Paper
161	CA-2016-162733	11 May 2016	12 May 2016	First Class	TT-21070	Consumer	United States	Los Angeles	California	West	OFF-PA-10002751	Office Supplies	Paper
172	CA-2014-118962	05 August 2014	09 August 2014	Standard Class	CS-12130	Consumer	United States	Los Angeles	California	West	OFF-PA-10000659	Office Supplies	Paper
173	CA-2014-118962	05 August 2014	09 August 2014	Standard Class	CS-12130	Consumer	United States	Los Angeles	California	West	OFF-PA-10001144	Office Supplies	Paper
559	CA-2016-113243	10 June 2016	15 June 2016	Standard Class	OT-18730	Consumer	United States	Los Angeles	California	West	OFF-PA-10003441	Office Supplies	Paper
694	CA-2015-131534	28 March 2015	02 April 2015	Standard Class	AB-10165	Consumer	United States	Los Angeles	California	West	OFF-PA-10000743	Office Supplies	Paper
761	CA-2015-132119	31 August 2015	05 September 2015	Standard Class	AR-10510	Consumer	United States	Los Angeles	California	West	OFF-PA-10003441	Office Supplies	Paper
904	CA-2016-130477	08 April 2016	12 April 2016	Standard Class	LC-11740	Consumer	United States	Los Angeles	California	West	OFF-PA-10002947	Office Supplies	Paper
906	CA-2016-130477	08 April 2016	12 April 2016	Standard Class	LC-11740	Consumer	United States	Los Angeles	California	West	OFF-PA-10000019	Office Supplies	Paper
1144	CA-2014-146969	29 September 2014	03 October 2014	Standard Class	AP-10915	Consumer	United States	Los Angeles	California	West	OFF-PA-10002105	Office Supplies	Paper
1569	US-2017-152002	11 June 2017	11 June 2017	Same Day	DV-13465	Consumer	United States	Los Angeles	California	West	OFF-PA-10000357	Office Supplies	Paper
1594	CA-2014-117317	19 October 2014	19 October 2014	Same Day	JF-15490	Consumer	United States	Los Angeles	California	West	OFF-PA-10004519	Office Supplies	Paper
1609	US-2016-115819	19 April 2016	24 April 2016	Second Class	JO-15280	Consumer	United States	Los Angeles	California	West	OFF-PA-10002377	Office Supplies	Paper
1683	US-2016-101497	29 September 2016	01 October 2016	Second Class	PS-18760	Consumer	United States	Los Angeles	California	West	OFF-PA-10000176	Office Supplies	Paper
1778	CA-2015-105347	24 November 2015	28 November 2015	Standard Class	DP-13000	Consumer	United States	Los Angeles	California	West	OFF-PA-10000675	Office Supplies	Paper
2083	CA-2017-104647	23 February 2017	01 March 2017	Standard Class	CK-12595	Consumer	United States	Los Angeles	California	West	OFF-PA-10002870	Office Supplies	Paper
2348	CA-2015-116092	15 February 2015	18 February 2015	Second Class	JM-16195	Consumer	United States	Los Angeles	California	West	OFF-PA-10004285	Office Supplies	Paper
2352	CA-2015-116092	15 February 2015	18 February 2015	Second Class	JM-16195	Consumer	United States	Los Angeles	California	West	OFF-PA-10000477	Office Supplies	Paper
2413	CA-2014-151295	12 November 2014	16 November 2014	Standard Class	JA-15970	Consumer	United States	Los Angeles	California	West	OFF-PA-10001947	Office Supplies	Paper
2492	CA-2014-113579	12 December 2014	15 December 2014	Second Class	KD-16345	Consumer	United States	Los Angeles	California	West	OFF-PA-10001457	Office Supplies	Paper
2837	CA-2017-110842	11 November 2017	16 November 2017	Standard Class	GA-14725	Consumer	United States	Los Angeles	California	West	OFF-PA-10000520	Office Supplies	Paper
2888	CA-2017-102099	18 December 2017	19 December 2017	First Class	EP-13915	Consumer	United States	Los Angeles	California	West	OFF-PA-10003172	Office Supplies	Paper
2889	CA-2017-102099	18 December 2017	19 December 2017	First Class	EP-13915	Consumer	United States	Los Angeles	California	West	OFF-PA-10000289	Office Supplies	Paper
3422	CA-2016-134803	28 May 2016	03 June 2016	Standard Class	CL-12565	Consumer	United States	Los Angeles	California	West	OFF-PA-10003790	Office Supplies	Paper
3557	CA-2017-118542	01 December 2017	05 December 2017	Standard Class	CC-12550	Consumer	United States	Los Angeles	California	West	OFF-PA-100004735	Office Supplies	Paper
3565	CA-2016-130029	03 July 2016	06 July 2016	First Class	GT-14755	Consumer	United States	Los Angeles	California	West	OFF-PA-10000552	Office Supplies	Paper
3945	CA-2016-133144	16 May 2016	21 May 2016	Second Class	DO-13435	Consumer	United States	Los Angeles	California	West	OFF-PA-10004971	Office Supplies	Paper

Step 2: Cleaning the Data

Once the data was in, I opened Power Query Editor to clean it up before using it anywhere. Here's what I did:

- Used the first row of the file as column headers, since they weren't set properly by default.
- Changed the data types so Sales, Quantity, and Discount were read as numbers, not text.
- Fixed the Order Date and Ship Date columns by setting the correct locale, so the dates loaded properly instead of turning into errors.
- Checked the column quality panel to make sure important columns like Row ID, Order ID, Customer ID, and Segment had no empty or error values.
- Also had a quick look at the distinct and unique value counts for each column, just to understand things like how many different customers and products were in the data.
- Renamed the query to something clearer so it would be easy to identify later in the model.



After this, every key column showed 100% valid values, so the table was ready to build on. Doing this cleaning step properly mattered a lot, because any errors here would have carried over into every chart and measure built later.

Step 3: Building Columns and Measures

With clean data, I moved to the modeling part. I added a Month column using the FORMAT function on Order Date, so I could see sales by month name instead of the full date. I did the same for Month Number, Quarter, and Year, which made it much easier to sort and group things properly.

I also built a few measures to use across the report:

- Total Sales — adds up all the sales values.
- Total Profit — adds up all the profit values.
- Total Quantity — total number of units sold.
- Total Orders — counts how many distinct orders were placed.

Having these as measures instead of just numbers pasted somewhere means they update automatically whenever a filter or slicer is applied, which is exactly what makes a Power BI report interactive.

The screenshot shows the Microsoft Power BI Desktop interface. The 'Table tools' ribbon is active, displaying options for 'Column tools' and 'Table tools'. The 'Column tools' section includes 'Name', 'Data type', 'Format', 'Summarization', 'Data category', 'Sort by column', 'Data groups', 'Manage relationships', and 'New column'. The 'Table tools' section includes 'Structure', 'Formatting', and 'Properties'. The main area displays a table with columns: Product Name, Sales, Quantity, Discount, Profit, Year, Quarter, Month, and Month Number. The table contains 12 distinct values for the 'Month' column. The 'Data' pane on the right shows a search bar and a list of fields including Category, City, Country, Customer ID, Discount, Month, Month Number, Order Date, Order Date Hierarchy, Order ID, Product ID, Product Name, Profit, Profit Margin, Quantity, Quarter, Region, Row ID, Sales, Segment, Ship Date, and Ship Mode.

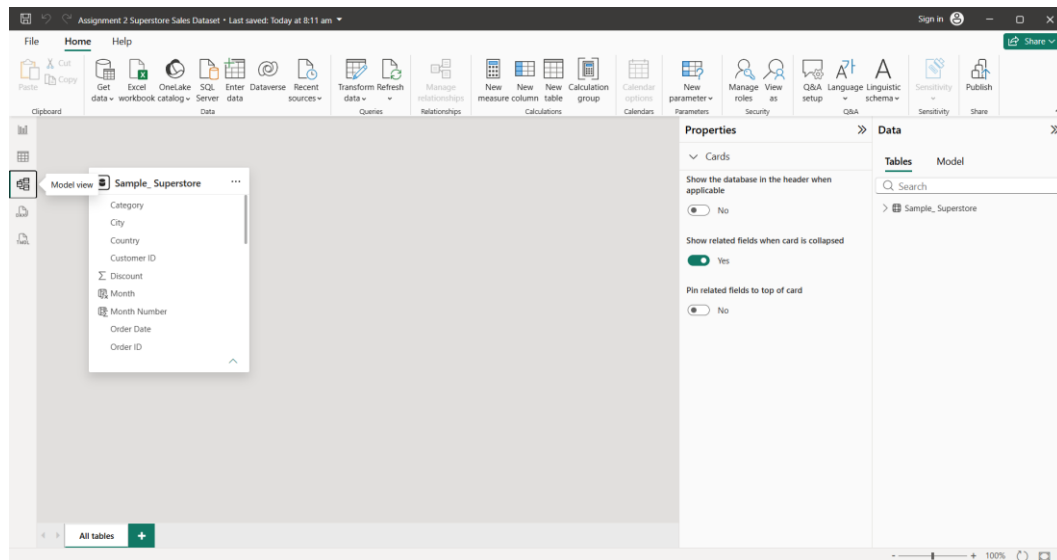
Product Name	Sales	Quantity	Discount	Profit	Year	Quarter	Month	Month Number
Bevis Oval Conference Table, Walnut	913.43	5	0.3	-169.637	2017	Q1	Jan	1
Samsung Galaxy S4	2003.168	4	0.2	250.396	2017	Q1	Jan	1
#10 White Business Envelopes, 4 1/8 x 9 1/2	125.36	8	0	58.9192	2016	Q1	Jan	1
AT&T 1080 Phone	109.592	1	0.2	8.2194	2016	Q1	Jan	1
SimpliFile Personal File, Black Granite, 15w x 6-15/16d x 11-1/4h	18.16	2	0.2	1.816	2017	Q1	Jan	1
Tennisco Single-Tier Lockers	1126.02	3	0	56.301	2017	Q1	Jan	1
Chromcraft Round Conference Tables	313.722	3	0.4	-99.3453	2016	Q1	Jan	1
Newell 317	7.056	3	0.2	0.7938	2015	Q1	Jan	1
Bagged Rubber Bands	2.52	2	0	0.1008	2017	Q1	Jan	1
Jensen SMP5-640 - speaker phone	1103.352	3	0.2	8.2764	2017	Q1	Jan	1
Panasonic KX-TG9541B DECT 6.0 Digital 2-Line Expandable Cordless Phone With Digital Answering Sys	359.976	3	0.2	130.4913	2017	Q1	Jan	1
DAX Charcoal/Nickel-Tone Document Frame, 5 x 7	18.96	2	0	8.532	2017	Q1	Jan	1
Bush Advantage Collection Racetrack Conference Table	1018.104	4	0.4	-373.3048	2015	Q1	Jan	1
Enemax Acrylux Wireless Keyboard	398.4	5	0.2	84.66	2015	Q1	Jan	1
Logitech 910-002974 M325 Wireless Mouse for Web Scrolling	47.984	2	0.2	14.3952	2017	Q1	Jan	1
Southworth 25% Cotton Granite Paper & Envelopes	6.54	1	0	3.0084	2014	Q1	Jan	1
Global Deluxe Stereo Chair	207.846	3	0.1	23.5872	2017	Q1	Jan	1
Novimex Swivel Fabric Task Chair	120.784	1	0.2	-13.5882	2017	Q1	Jan	1
Acco 6 Outlet Guardian Premium Surge Suppressor	87.36	6	0	23.5872	2015	Q1	Jan	1
Bagged Rubber Bands	7.56	6	0	0.3024	2016	Q1	Jan	1
Fellowes Mobile File Cart, Black	348.208	7	0.2	30.4682	2017	Q1	Jan	1
LG Electronics Tone+ HBS-730 Bluetooth Headset	107.118	3	0.4	-21.4236	2015	Q1	Jan	1
Anker 24W Portable Micro USB Car Charger	26.376	4	0.4	2.6376	2015	Q1	Jan	1
Staple envelope	23.36	2	0	11.68	2016	Q1	Jan	1
Xerox 1881	12.28	1	0	5.7716	2017	Q1	Jan	1
GE 30524EE4	235.188	2	0.4	-43.1178	2015	Q1	Jan	1
Xerox 195	40.08	6	0	19.2384	2014	Q1	Jan	1

Step 4: Setting Up the Model

Before building the report pages, I checked the model view to see all the fields available and confirmed the table and its properties were set up correctly, so everything downstream would work properly.

- Went through the list of fields in the Data pane to make sure every column and measure I needed was showing up correctly.

- Confirmed the card settings so related fields would be visible when needed, which helps when clicking through the model later.
- This step is a bit boring compared to building charts, but skipping it usually leads to confusing errors further down the line.



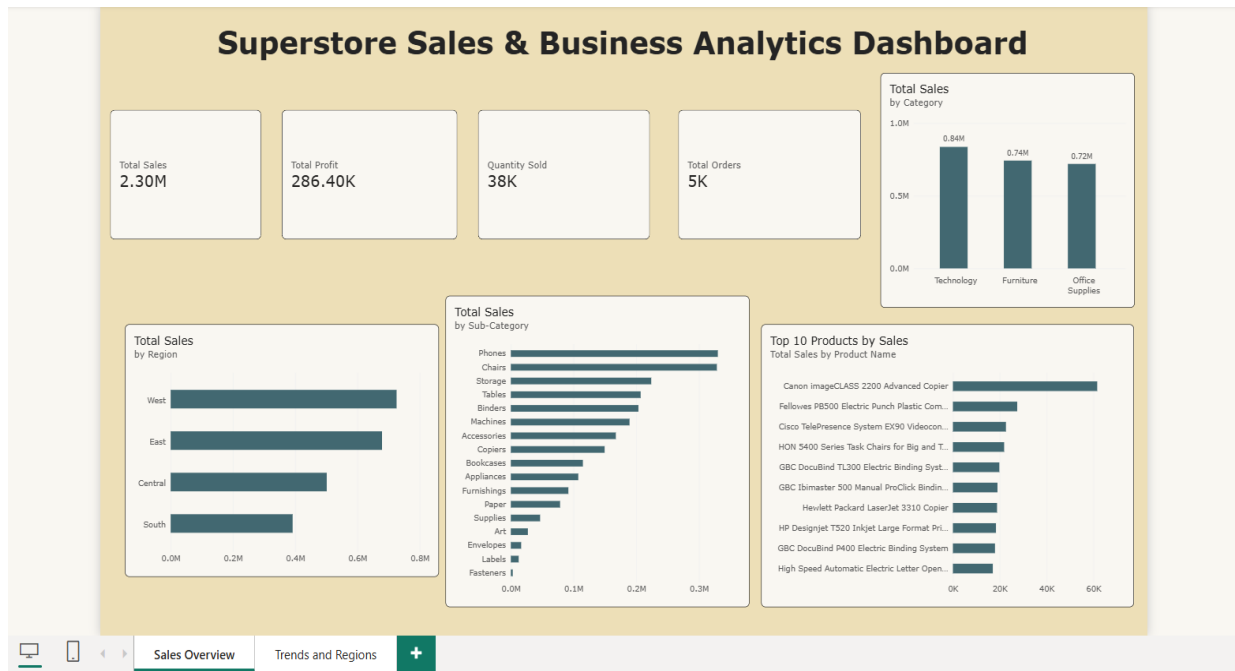
Step 5: Building the Dashboard

Page 1 — Sales Overview

The first page gives a quick snapshot of the business. At the top, I added four KPI cards: Total Sales (2.30M), Total Profit (286.40K), Quantity Sold (38K), and Total Orders (5K).

- A bar chart showing Total Sales by Category.
- A bar chart showing Total Sales by Region.
- A bar chart showing Total Sales by Sub-Category.

I kept this page fairly simple on purpose. The idea was that someone should be able to glance at it for ten seconds and already know how the business is doing, without needing to dig into anything else.

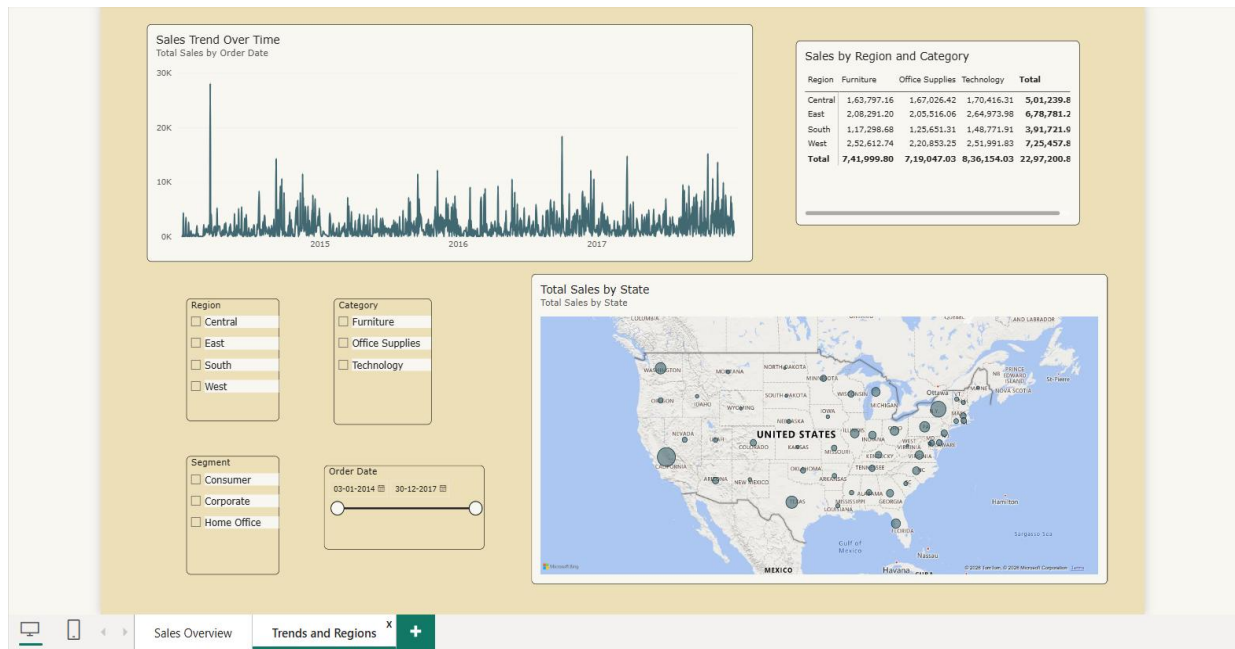


Page 2 — Trends and Regions

The second page looks more closely at trends over time and by location:

- A line chart showing Sales Trend Over Time, from 2014 to 2017.
- A table showing Sales by Region and Category, with totals.
- A map showing Total Sales by State, using bubble size.
- Slicers for Region, Category, Segment, and Order Date, so the page can be filtered on the fly.

This page is more for exploring than for a quick glance. Someone could use the slicers to check, say, just the West region's sales in 2016, or compare Technology against Furniture over time.



Key Insights

- Technology and Furniture bring in the most sales, followed closely by Office Supplies.
- West and East regions consistently sell more than Central and South.
- Phones, Chairs, and Storage are the top sub-categories; Labels and Fasteners barely contribute.
- Sales show clear spikes at certain points in the timeline, which is worth digging into further.
- Los Angeles shows up a lot in the raw data, suggesting California is one of the stronger markets overall.
- The Consumer segment appears to place orders more often than Corporate or Home Office, based on how frequently it shows up in the data.
- Standard Class is clearly the most common shipping method used across orders.
- Looking at sales by month makes it easier to plan ahead, since certain months consistently perform better than others.

Conclusion

This assignment gave me hands-on practice with the whole Power BI process — from importing and cleaning raw data, to adding calculated columns and DAX measures, to building an actual interactive dashboard. The final report gives a clear, filterable view of how Superstore is performing across categories, regions, and time, and it helped me understand how a plain spreadsheet can be turned into something genuinely useful for decision-making.

Overall, this was a good exercise in thinking like an analyst rather than just a student following steps. Cleaning the data properly, choosing the right calculations, and deciding what to actually show on a dashboard all mattered just as much as knowing which buttons to click in Power BI.

Learning Outcomes

- Learned how to import and clean data properly using Power Query before building anything on top of it.
- Got comfortable writing simple DAX formulas to create calculated columns and measures.
- Understood how KPI cards, bar charts, line charts, and maps each serve a different purpose on a dashboard.
- Practiced adding slicers to make a report interactive instead of just a static set of charts.
- Learned to think about who will actually use the dashboard, and design each page around what they'd want to know.